Questions:

Part I

Answer all the questions.

Choose the most suitable answer from the given four alternatives and write the option code with the corresponding answer.

1.The basic structural unit of silicates is ______.

(a) $(SiO_3)^{2-}$ (b) $(SiO_4)^{2-}$ (c) $(Sio)^{-}$ (d) $(SiO_4)^{4-}$

2.Which of the following oxidation states is most common among the lanthanoids?

(a) +4 (b) +2 (c) +5 (d) +3

3.A complex has a molecular formula $MSO_4Cl.6H_2O$. The aqueous solution of it gives white precipitate with Barium chloride solution and no precipitate is obtained when it is treated with silver nitrate solution. If the secondary valence of the metal is six, which one of the following correctly represents the complex?

(a) $[M(H_2O)_4Cl]SO_4.2H_2O$ (b) $[M(H_2O)_6]SO_4$ (c) $[M(H_2O)_5Cl]SO_4.H_2O$ (d) $[M(H_2O)_3Cl]SO_4.3H_2O$

4.For a first order reaction A ⟶ B the rate constant is x min^{-1}. If the initial concentration of A is 0.01M, the concentration of A after one hour is given by the expression.

(a) 001. e^{-x} (b) 1 x $10^{-2}(1-e^{-60x})$ (c) $(1 \times 10^{-2})e^{-60x}$ (d) none of these

5.Which metal is used for extraction of Au and Ag and also for galvanisation of iron objects?

(a) Mg (b) Zn (c) Cr (d) Co

6.In a solid, the cation is missing from the lattice and is found in an interstitial position. The defect is _______.

(a) Metal excess (b) p type (c) Schottky (d) Frenkel

7.The hybridisation and shape of SF_6 is respectively?

(a) sp^3d^2, square planar (b) sp^3d^2, octahe[illegible] [illegible]) sp^3d see-saw (d) sp^3d, trigonal bipyramidal

(a) Grignard reagent (b) Sn / HCl Chemistry
(c) hydrazine in presence of slightly acidic solution (d) hydrocyanic acid

11.Which among the following reagent is not used to differentiate ethanol and phenol?

(a) neutral $FeCl_3$ (b) $C_6H_5N_2Cl$ (c) NaOH (d) anhy $ZnCl_2$

12.The solution whose pH is maintained constant even upon the addition of small amounts of acid or base is called _______.

(a) acidic solution (b) basic solution (c) buffer solution (d) true solution

13.The process in which chemical change occurs on passing electricity is termed as _____.

(a) neutralisation (b) hydrolysis (c) electrolysis (d) ionisation

14.Pick out the odd one.

(a) Wax (b) Starch (c) Glucose (d) Fructose

15.Which among the following exists as zwitter ion?

(a) sulphanilic acid (b) salicylic acid (c) acetanilide (d) all the above

Part II

Answer any 6 questions. Question no. 24 is compulsory.

16.What is catenation ? describe briefly the catenation property of carbon.

17.Write a note on zeolites.

18.What are point defects?

19.Give the geometry and magnetic character of $[NiCl_4]^{2-}$

20.Write the two similarities between calcination and roasting. Chemicals Industry

21.Discuss the oxidising power of fluorine.

22.Why is instantaneous rate preferred over average rate?

23.The concentration of hydroxide ion in a water sample is found to be 2.5×10^{-6}M. Identify the nature of the solution.

24.Write the expression for the solubility product of Hg_2Cl_2 .

25.Define conductance. Give its unit

Part III

Answer any 6 questions. Question no. 33 is compulsory.

26.The half life of the homogeneous gaseous reaction $SO_2Cl_2 \rightarrow SO_2 + Cl_2$ which obeys first order kinetics is 8.0 minutes. How long will it take for the concentration of SO_2Cl_2 to be reduced to 1% of the initial value?

27.Explain alkali leaching in the extraction of aluminum.

28.Complete the following reactions?
i) $Cr^{2} + 2e^{-} \longrightarrow$
ii) $Mn^{2+} + 2e^{-} \longrightarrow$
iii) $Fe^{2+} + 2e^{-} \longrightarrow$
iv) $CO^{2+} + 2e^{-} \longrightarrow$

29.Can we use nucelophiles such as NH_3,CH_3O^- for the Nucleophilic substitution of alcohols

30.Give chemical tests to distinguish between propan-2-ol and 2-methyl-propan-2-ol.

31.Write the reactions taking place in anode and cathode of a mercury button cell. Give the over all redox reaction of the cell with the emf generation

32.Amino acids are amphoteric in nature. Explain.

33.Write a short note on Antioxidants.

34.How can the following conversion be effected?

i) Nitrobenzene to anisole

ii) Chloro benzene to phenyl hydrazine

iii) Aniline to benzoic acid

iv) Benzene diazonium chloride to ehtyl benzene.

Part IV

Answer all the questions.

35.Indicate the types of isomerism exhibited by the following complexes and draw the structures for these isomers.

(i) $K[Cr(H_2O)_2 (C_2O_4)_2]$

(ii) $[Co(en)_3]Cl_3$

(iii) $[Co(NH_3)_5(NO_2)](NO_3)_2$

(iv) $[Pt (NH_3)(H_2O)Cl_2]$

36.What are the general characteristics of solids?

37.Account for the following:

(i) Reducing character decreases from SO_2 to TeO_2.

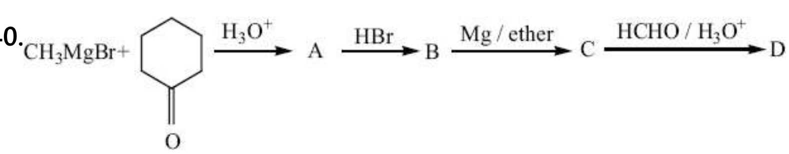
(ii) Xenon forms compounds with fluorine and oxygen only.

38.Justify the following statement.

"Elements of the first transition series possess many properties different from those of heavier transition elements".

39.Describe some feature of catalysis by Zeolites.

40.



Identify A,B,C,D and write the complete equation

41.Identify A, B, C and D

$$\text{ethanoic acid} \xrightarrow{SOCl_2} A \xrightarrow{Pd/BaSO_4} B \xrightarrow{NaOH} C \xrightarrow[\Delta]{} D$$

42.Explain the electrical property of colloids with a neat diagram. (or) Write a note on Helmholta electrical double layer.

43.Compound (A) of molecular formula C_7H_8 when treated with air in presence of V_2O_5 at 773 K gives a compound (B) of molecular formula C_7H_6O, which has the smell of bitter almonds. Alkaline $KMnO_4$ oxidised compound (B) to (C) of molecular formula C_7H_6Or Compound (B) on treatment with N_2H_4 and KOH gives back compound (A). Identify (A), (B) & (C) and explain the reactions.

Answers Key:

Part I

Answer all the questions.

Choose the most suitable answer from the given four alternatives and write the option code with the corresponding answer.

1. (d) $(SiO_4)^{4-}$
2. (d) +3
3. Molecular formula: $MSO_4Cl.6H_2O$

   Formation of white precipitate with Barium chloride indicates that SO^{2-}_4 ions are outside the coordination sphere, and no precipitate with $AgNO_3$ solution indicates that the Cl^- ions are inside the coordination sphere. Since the coordination number of M is 6.Cl^- and 5 H_2O are ligands, remaining 1 H_2O molecular and $SO^{2-}{}_4$ are in the outer coordination sphere.

4. $k = \frac{2.303}{t} log \frac{[A_0]}{[A]}$

   $k = \frac{1}{t} ln \frac{[A_0]}{[A]}$

   $e^{kt} = \frac{[A_0]}{[A]}$

   $[A] = [A_0]\ e^{kt}$

   In this case

   $k = x\ min^{-1}$ and $[A_0] = 0.01\ M = 1 \times 10^{-2} M$

   t = 1 hour = 60 min

   $[A] = (1 \times 10^{-2}) e^{-60x}$
5. (b) Zn
6. (d) Frenkel
7. (b) sp^3d^2, octahedral
8. $PbI_{2(s)} \rightleftharpoons Pb^{2+}{}_{(aq)} + 2I^-{}_{(aq)}$

   $K_{sp} = (s)\ (2s)^2$

   $3.2 \times 10^{-8} = 4s^3$

   $s = (3.2 \times 10^{-8}/4)^{1/3}$

   $= (8 \times 10^{-9})^{1/3}$

   $= 2 \times 10^{-3}$ M
9. dispersion medium-gas

   dispersed phase-liquid
10. (c) hydrazine in presence of slightly acidic solution
11. (d) anhy $ZnCl_2$
12. (c) buffer solution
13. (c) electrolysis
14. (a) Wax
15. (a) sulphanilic acid

Part II

Answer any 6 questions. Question no. 24 is compulsory.

16. Catenation is an ability of an element to form chain of atoms.

    The conditions for catenation.

    (a) The valency of element is greater than or equal to two.

    (b) Element should have an ability to bond with itself

    (c) The self bond must be as strong as Its bond with other elements

    (d) Kinetic inertness of catenated compound towards other molecules.

    (e) Carbon possesses all the above properties and forms a wide range of compounds with itself and with other elements such as H, O, N, S and halogens.
17. (i) Zeolites are three-dimensional crystalline solids containing Al, Si and O in their regular three dimensional framework.

    (ii) They are hydrated sodium alumino silicates with general formula $Na_2O(Al_2O_3).x(SiO_2)\cdot yH_2O$ (x = 2 to 10; y = 2 to 6).

    (iii) Zeolites have porous structure in which the monovalent sodium ions and water molecules are loosely held.

    (iv) The Si and Al atoms are tetrahedrally coordinated with each other through shared oxygen atoms.

(v) Zeolites are similar to clay minerals but they differ in their crystalline structure.

(vi) Zeolites have a three dimensional crystalline structure looks like a honeycomb consisting of a network of interconnected tunnels and cages.

(vii) Water molecules moves freely in and out of these pores but the zeolite framework remains rigid

(viii) Another special aspect of this structure is that the pore/channel sizes are nearly uniform, allowing the crystal to act as a molecular sieve.

18. The imperfection occurs due to missing atoms, displaced atoms or extra atoms, is named as a point defect. Such defects arise due to imperfect packing during the original crystallisation or they may arise from thermal vibrations of atoms at elevated temperatures.
19. Tetrahedral, paramagnetic.
20. (i) The end product of both the processes is oxide of metal.
    (ii) Volatile impurities are removed from the ore and surface area for the further reaction increases.
21. (i) All the halogens have strong oxidising property due to their high electron affinity.
    (ii) Fluorine is the strongest oxidising agent. It oxidises other halide ions into halogens in solution or when dry.
    $F_2 + 2X^- \longrightarrow 2F^- + X_2$ (where X$^-$ = Cl$^-$, Br$^-$, I$^-$)
22. Rate decreases with time as the reaction proceeds and the average rate cannot be used to predict the rate of the reaction at any instant. The rate of the reaction, at a particular instant during the reaction is called the instantaneous rate. So instantaneous rate in prepared over average rate.
23. 1. If $\left[\mathrm{OH}^-\right] > 1 \times 10^{-7}\mathrm{M}$, the solution is basic. $2.5 \times 10^{-6}\mathrm{M} > 1 \times 10^{-7}\mathrm{M}$
    2. $\therefore$ The solution is basic.
24. $\mathrm{Hg_2Cl_{2(s)}} \rightleftharpoons \mathrm{Hg_2^{2+}}\ \mathrm{(aq)}\ + 2\mathrm{Cl^-}_{\mathrm{(aq)}} s \qquad s \qquad 2s$
    $\mathrm{K_{sp}} = \left[\mathrm{Hg_2^{2+}}\right]\left[\mathrm{Cl^-}\right]^2$
    $= (\mathrm{s})(2\ \mathrm{s})^2$
    $\mathrm{K_{sp}} = 4\ \mathrm{s}^3$
25. The reciprocal of the resistance $\left(\frac{l}{R}\right)$ gives the conductance of an electrolytic solution. The SI unit of conductance is Siemen (S).

Part III

Answer any 6 questions. Question no. 33 is compulsory.

26. $\mathrm{k} = 0.693/\mathrm{t}_{1/2}$
    $\mathrm{k} = \frac{0.693}{8.0} = 0.08\ \mathrm{min}^{-1}$
    For the first order reaction:
    $t = \frac{2.303}{k} \log \frac{[A_0]}{[A]}$
    $t = \frac{2.303}{0.087} \log\left(\frac{100}{1}\right) = 26.47 \log 10^2$
    $t = 2 \times 26.47 \log 10$
    $t = 52.94\ \mathrm{min}$
27. (i) In this method, the ore is treated with aqueous alkali to form a soluble complex.
    (ii) Bauxite, an important ore of aluminum is heated with a solution of sodium hydroxide or sodium carbonate in the temperature range 470 - 520 K at 35 atm to form soluble sodium meta-aluminate leaving behind the impurities, iron oxide and titanium oxide.

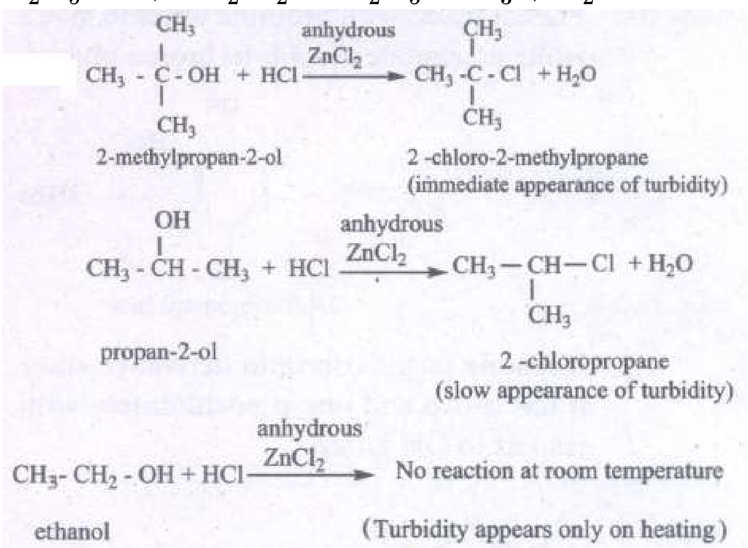
$$Al_2O_{3(s)} + 2NaOH_{(aq)} + 3H_2O_{(l)} \longrightarrow 2Na[Al(OH)_4]_{(aq)}$$

(iii) The hot solution is decanted, cooled, and diluted. This solution is neutralised by passing CO_2 gas, to the form hydrated Al_2O_3 precipitate

$$2Na[Al(OH)_4]_{(aq)} + CO_{2(g)} \longrightarrow Al_2O_3.xH_2O_{(s)} + 2NaHCO_{3(aq)}$$

(iii) The precipitate is filtered off and heated around 1670 K to get pure alumina Al_2O_3

28. i) $Cr^2 + 2e^- \longrightarrow Cr$
    ii) $Mn^{2+} + 2e^- \longrightarrow Mn$
    iii) $Fe^{2+} + 2e^- \longrightarrow Fe$
    iv) $CO^{2+} + 2e^- \longrightarrow Co$
29. Yes, we can use nucleophiles such as NH_3, CH_3O^- for the nucleophilic substitution of alcohol.
    Example
    $$C_2H_5OH + NH_3 \rightarrow C_2H_5NH_2 + H_2O$$
    $$C_2H_5OH + CH_2\,N_2 \rightarrow C_2H_5OCH_3 + N_2$$

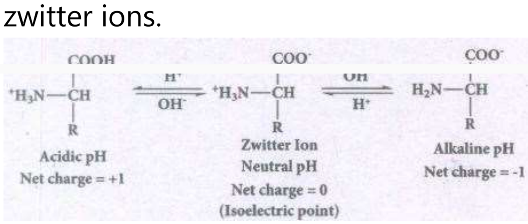


30.

31. (i) Oxidation occurs at anode:
    $$\overset{0}{Zn}_{(s)} + \underset{\text{from KOH}}{2OH^-_{(aq)}} \longrightarrow \overset{+2}{Zn}O_{(s)} + H_2O(l) + 2e^-$$

    (ii) Reduction occurs at cathode:
    $$\overset{+2}{Hg}O_{(s)} + H_2O(l) + 2e^- \longrightarrow \overset{0}{Hg}(l) + 2OH^-_{(aq)}$$

    (iii) Overall reaction:
    $$Zn_{(s)} + HgO_{(s)} \rightarrow ZnO_{(s)} + Hg(l)$$
    (iv) Cell emf: about 1.35V.
32. At aqueous solution, the proton from carboxyl group can be transferred to the amino group of an amino acid leaving these groups with opposite charges. Despite having both positive and negative charges, this molecule is neutral and has amphoteric behaviour. These ions are called zwitter ions.



Zwitter ions behave as neutral molecules at pH isoelectric point.

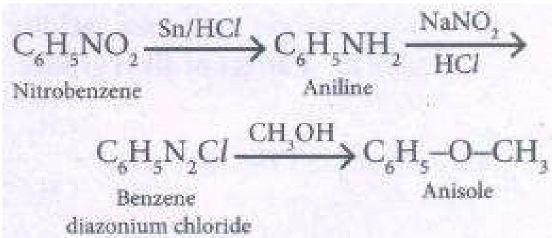
33. (i) Antioxidants are substances which retard the oxidative deteriorations of food.

(ii) Food containing fats and oils is easily oxidised and turn rancid.

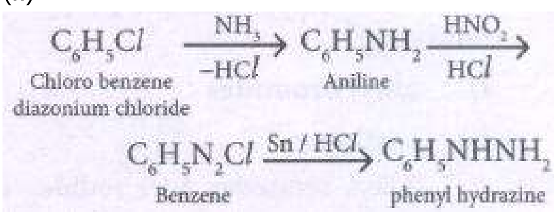
(iii) To prevent the oxidation of the fats and oils, chemical BHT(butylhydroxy toluene), BHA(Butylated hydroxy anisole) are added as food additives.

(iv) They are generally called antioxidants. These materials readily undergo oxidation by reacting with free radicals generated by the oxidation of oils, thereby stop the chain reaction of oxidation of food.

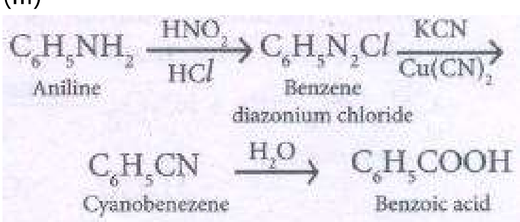
34. (i)



(ii)



(iii)



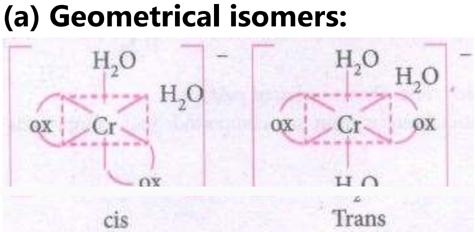
(iv)

Part IV

Answer all the questions.

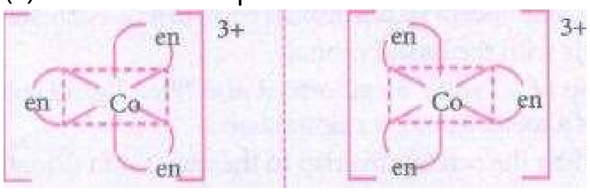
35. (i) It exhibits both geometrical and optical isomerism

(a) Geometrical isomers:



(b) Optical isomers:

(ii) It shows two optical isomers

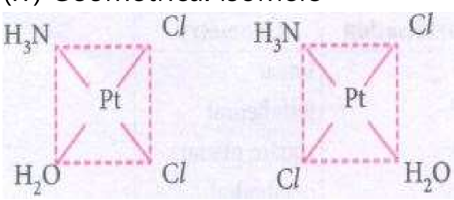


(iii) Ionisation isomers

Linkage isomers

$[Co(NH_3)_5(NO_2)(NO_3)_2]$, $[CO(NH_3)_5\ (ONO)](NO_3)_2$

(iv) Geometrical isomers



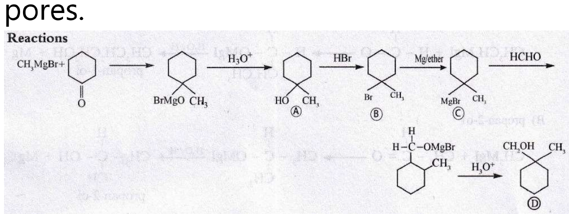
36. (i) Solids have definite volume and shape
    (ii) Solids are rigid and incompressible
    (iii) Solids have strong cohesive forces.
    (iv) Their constituents have fixed positions and can only oscillate about their mean positions.
37. (i) Stability of higher oxidation state decreases down the group from S to Te because stability of lower oxidation state increases down the group from S to Te.
    (ii) Fluorine and oxygen are most electronegative and very reactive. So they can ionised noble gases and form compounds.
38. The heavier transition elements belong to fourth (4d), fifth (Sd) and sixth (6d) transition series. Their properties are expected to be different form the elements belonging to the first (3d) series due to the following reasons.
    (i) Atomic radii: Size of the transition elements 94d and Sd series are larger than those of the corresponding elements of the first transition series though those of 4d and Sd series are very close to each other.
    (ii) Ionisation enthalpy of Sd series are higher than the corresponding elements of 3d and 4d series.
    (iii) Atomisation enthalpy of 4d and Sd series are higher than the corresponding elements of the first series.
    (iv) Melting and boiling points of heavier transition elements are greater than those of the first transition series due to stronger intermetallic bonding.
    (v) The elements of the first transition series generally form low or high spin complexes, depending upon the higher of ligand field. However, the heavier transition elements form low spin complexes irrespective of the strength of the ligand filed.
39. (i) Zeolites are microporous, crystalline, hydrated, alumino silicates, made of silicon and aluminium tetrahedra.

(ii) There are about 50 natural zeolites and 150 synthetic zeolites.
(iii) As silicon is tetravalent and aluminium is trivalent, the zeolite matrix carries extra negative charge.
(iv) To balance the negative charge, there are extra framework cations for example H^+or Na^+ons. Zeolites carrying protons are used as solid acids, catalysis and they are extensively used in the petrochemical industry for cracking heavy hydrocarbon fractions into gasoline, diesel, etc.,
(v) Zeolites carrying Na^+ ions are used as basic catalysis.
(vi) One of the most important applications of zeolites is their shape selectivity.
(vii) In zeolites, the active sites namely protons are lying inside their pores. So, reactions occur only inside the pores of zeolites.

Reactant selectivity:
When bulkier molecules in a reactant mixture are prevented from reaching the active sites within the zeolite crystal, this selectivity is called reactant shape selectivity.

Transition state selectivity:
If the transition state of a reaction is large compared to the pore size of the zeolite, then no product will be formed.

Product selectivity:
It is encountered when certain product molecules one too big to diffuse out of the zeolite pores.

40. 

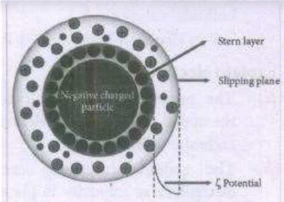
$$CH_3COOH \xrightarrow{SOCl_2} \underset{\text{Acetylchloride}}{\underset{(A)}{CH_3COCl}} \xrightarrow{pd/BaSO_4} \underset{\text{Acetaldehyde}}{\underset{(B)}{CH_3-CHO}}$$

$$\underset{\text{Acetaldehyde}}{\underset{(B)}{CH_3-CHO}} \xrightarrow{NaOH} \underset{\text{aldol}}{\underset{(C)}{CH_3-\underset{OH}{\underset{|}{CH}}-CH_2-CHO}} \xrightarrow{\Delta} \underset{\text{Crotonaldehyde}}{\underset{(D)}{CH_3-CH=CH-CHO}}$$

41.

Compound	**Name**
A	Acetylchloride
B	Acetaldehyde
C	3 - Hydroxy butanal
D	Crotonaldehyde

42. **Helmholtz double layer:**
(i) The surface of colloidal particle adsorbs one type of ion due to preferential adsorption.
(ii) This layer attracts the oppositely charged < ions in the medium and hence at the boundary separating the two electrical double layers are setup.
(iii) This is called as Helmholtz electrical double layer.
(iv) As the particies nearby are having similar: charges, they cannot come close and condense.



43. (i) Compound A is Toluene ($C_6H_5CH_3$).

(ii) When toluene is treated with air in the presence of V_2O_5 at 773K. It gives compound B benzaldehyde (C_6H_5CHO).

$$C_6H_5CH_3 \underset{air/v_2O_5 \quad 773k}{\xrightarrow{(O)}} C_6H_5CHO + H_2O$$

(iii) Alkaline $KMnO_4$ oxidises Benzaldehyde (B) to Benzoic acid (C).

$$C_6H_5CHO \xrightarrow{(O)} C_6H_5COOH \quad (B) \qquad\qquad (C)$$

(iv) Benzaldehyde (B) on treatment with N_2H_4 and KOH gives compound A toluene.

$$\underset{(B)}{C_6H_5CHO} \underset{N_2H_4/KOH}{\longrightarrow} \underset{(C)}{C_6H_5CH_3}$$

Compound	**Compound Name**	**Formula**
A	Toluene	$C_6H_5CH_3$
B	Benzaldehyde	C_6H_5CHO
C	Benzoic acid	C_6H_5COOH

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